Weekly Progress Report

Week 9: Physical Robot Arm Testing & EEG-Based Control Integration

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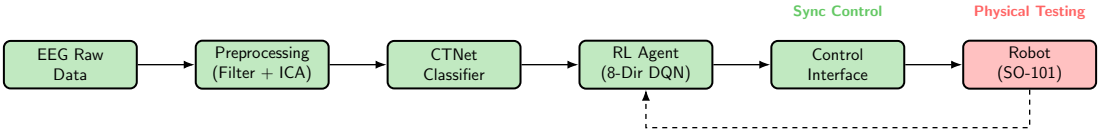
BCI Control System Project

March 4, 2026

System Overview: Current Focus



Legend:

- Completed
- In Progress
- **This Week's Focus**
- Pending

This Week's Objectives

Goals

1. Test 8-direction movement patterns on physical SO-101 robot arm
2. Implement synchronized joint movement for smooth homing/return
3. Integrate EEG subject sequences with physical robot control
4. Compare simulated vs physical robot trajectories

Key Question to Address

Can the physical robot arm accurately reproduce the movement patterns designed in simulation, and how does EEG-based control perform on real hardware?

Methods & Implementation

Synchronized Joint Control:

- All joints arrive at target simultaneously
- Velocity calculated based on distance
- Velocity boost for loaded joints (1.3x-1.5x)

Velocity Synchronization:

$$v_j = \frac{d_j}{t_{total}} \times \text{boost}_j$$

Implementation Files:

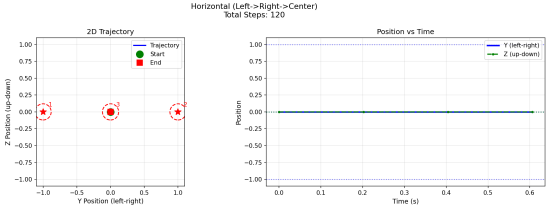
- `test_8dir_comprehensive.py`
- `eeg_physical_control.py`
- `serial_go_home_sync.py`
- `serial_go_return_sync.py`

Test Patterns (6 types):

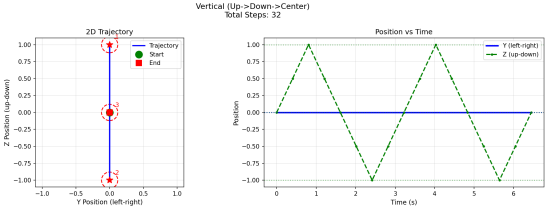
- Horizontal, Vertical
- Diagonal (2 patterns)
- Square CW/CCW

Physical Robot Test: Horizontal & Vertical Movement

Horizontal Movement:



Vertical Movement:

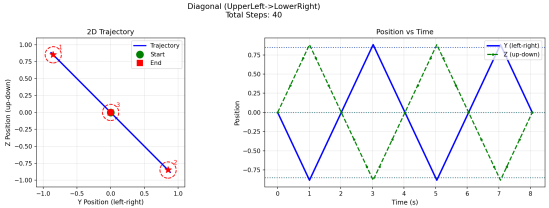


Observations:

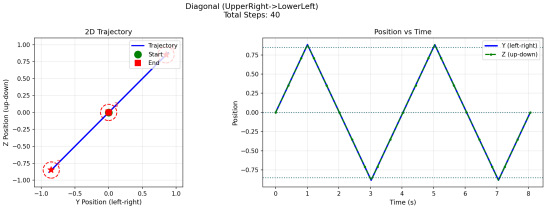
- Smooth trajectory with consistent velocity
- Y/Z positions tracked accurately to targets

Physical Robot Test: Diagonal Movement

Diagonal UL-DR:



Diagonal UR-DL:

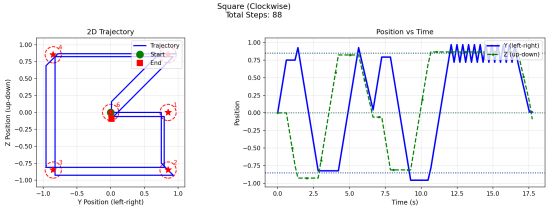


Observations:

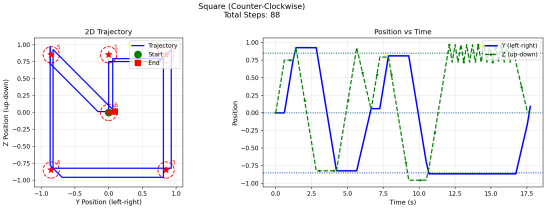
- Both Y and Z change simultaneously (true diagonal motion)
- 8-direction action space enables efficient diagonal paths

Physical Robot Test: Square Movement Patterns

Square Clockwise:



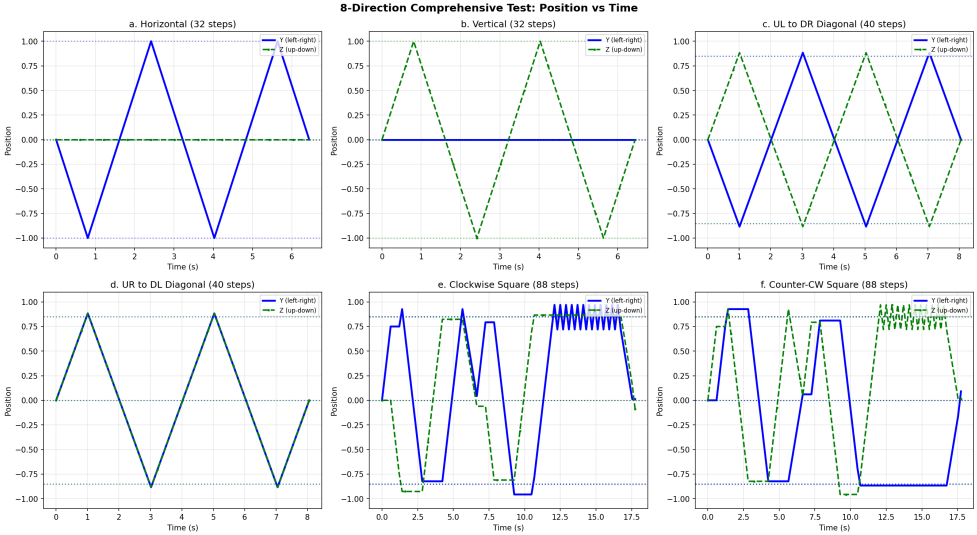
Square Counter-Clockwise:



Observations:

- Complex multi-target sequences completed successfully
- All 4 corners reached with good precision

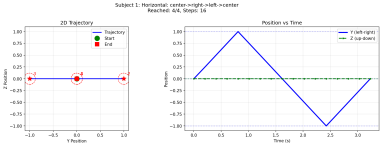
Physical Robot Test: All Patterns Overview



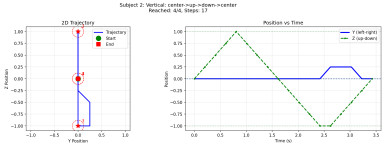
Summary: All 6 movement patterns executed successfully on physical robot

EEG-Based Physical Control: Subjects 1-3

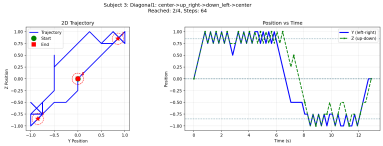
Subject 1 (Horizontal):



Subject 2 (Vertical):



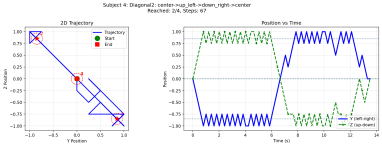
Subject 3 (Diagonal):



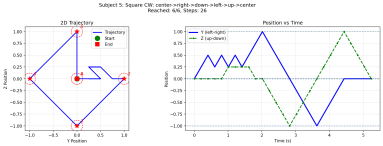
Note: EEG classification includes 10% noise to simulate real-world conditions

EEG-Based Physical Control: Subjects 4-6

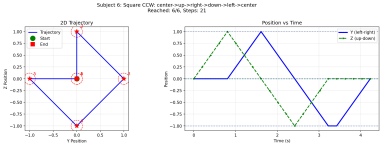
Subject 4 (Diagonal 2):



Subject 5 (Square CW):

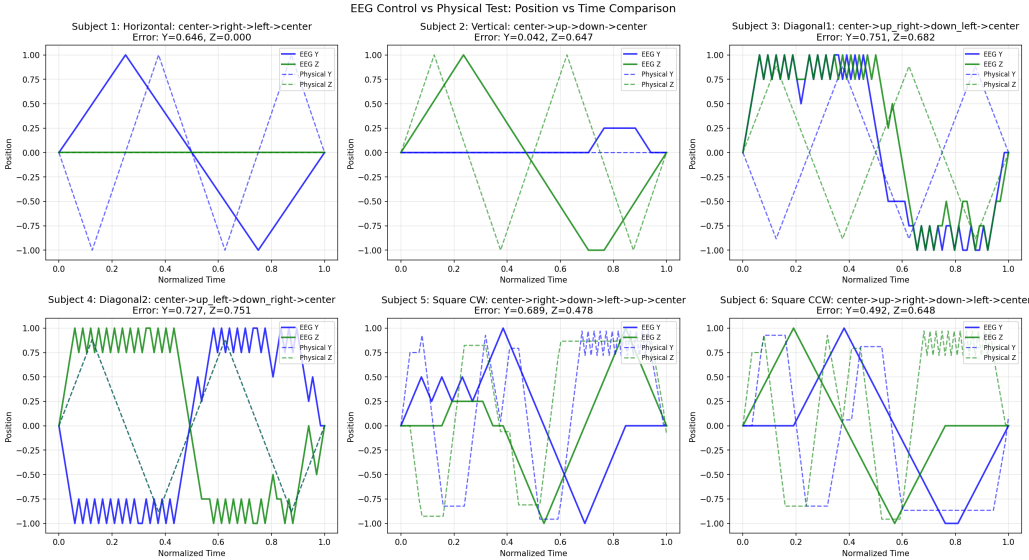


Subject 6 (Square CCW):



Note: Complex patterns (square) require more steps due to multiple targets

Comparison: EEG Control vs Physical Test Baseline



Error Analysis: EEG Control — Endpoint Error

Metric: Endpoint Error $d = \sqrt{(\Delta y)^2 + (\Delta z)^2}$

Subject	Error	Workspace%
S1–S2 (Horiz/Vert)	0.000	0.00%
S3–S4 (Diagonal)	0.107	5.36%
S5–S6 (Square)	0.000	0.00%
Mean	**0.036**	**1.79%**

Error Unit

All errors: **endpoint error** (final distance to target) in normalized coordinates $[-1, 1]$.
Workspace% $= \frac{\text{error}}{2.0} \times 100\%$. V1 step: 0.12 norm $= 0.35$ rad (20.1°). Reach threshold: < 0.12 (6%).

Challenges & Solutions

Challenges Encountered

- Joints not arriving simultaneously
- Wrist Flex slower due to load
- GOAL_TIME conflicts (500ms vs 4000ms)
- Torque disabled causing drift

Solutions Applied

- Synchronized velocity calculation
- Velocity boost (1.3x-1.5x) for loaded joints
- Unified move_time_ms = 2000ms
- Keep torque enabled during close()

Servo Velocity Compensation:

Joint	Base Velocity	Boosted Velocity
Shoulder Lift	300	390 (x1.3)
Wrist Flex	206	309 (x1.5)

Results Summary

Physical Robot Testing:

Metric	Value
Patterns Tested	6
Target Reach Rate	100%
Motion Smoothness	High
Joint Sync Error	<30 ticks

EEG-Based Control:

Metric	Value
Subjects Tested	6
Noise Level	10%
Mean Endpoint Error	0.036 (1.79%)
Target Reach Rate	81.8%

Key Achievement

Successfully demonstrated end-to-end BCI control pipeline from EEG simulation to physical robot movement with synchronized multi-joint control.

Next Week's Plan

Planned Tasks

1. Finalize report with all experimental results
2. Generate performance comparison tables
3. Create demonstration video of physical robot
4. Complete project documentation

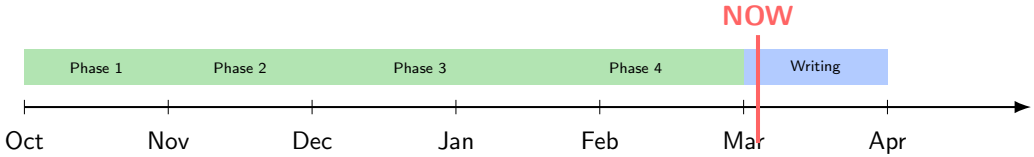
Expected Deliverables:

- Final project report (draft)
- Demo video with all 6 patterns
- Performance metrics summary

Project Status:

- All core components complete
- Physical testing validated
- Focus shifting to documentation

Project Timeline



Current Status: On track - All technical work complete, entering final documentation phase